

CBSE BOARD EXAMINATION 2023

MATHEMATICS (STANDARD)

Class 10th — Solved Paper with Answers

Maximum Marks:	80
Time Allowed:	3 Hours
Date of Exam:	March 21, 2023
Total Questions:	38 (All Compulsory)
Sections:	A (MCQs), B (VSA), C (SA), D (LA), E (Case Study)

GENERAL INSTRUCTIONS:

- (i) This question paper contains 38 questions. All questions are compulsory.
- (ii) This question paper is divided into five sections — A, B, C, D and E.
- (iii) Section A: Q1–Q18 are MCQs (1 mark each); Q19–Q20 are Assertion-Reason (1 mark each).
- (iv) Section B: Q21–Q25 are Very Short Answer type questions (2 marks each).
- (v) Section C: Q26–Q31 are Short Answer type questions (3 marks each).
- (vi) Section D: Q32–Q35 are Long Answer type questions (5 marks each).
- (vii) Section E: Q36–Q38 are Case Study based questions (4 marks each).
- (viii) There is no overall choice. Internal choice is provided in some questions.
- (ix) Draw neat diagrams wherever required. Take $\pi = 22/7$ wherever required.
- (x) Use of calculators is NOT allowed.

SECTION A — MULTIPLE CHOICE QUESTIONS (1 Mark Each)

Q1. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$, where x, y are prime numbers, then $\text{HCF}(a, b)$ is

- (A) xy
- (B) xy^2
- (C) x^3y^3
- (D) x^2y^2

Answer: (B)

Solution: HCF = product of lowest powers of common prime factors = $x^1 \times y^2 = xy^2$.

Q2. The LCM of the smallest two-digit composite number and the smallest composite number is

- (A) 12
- (B) 4
- (C) 20
- (D) 44

Answer: (C)

Solution: Smallest two-digit composite = 10, smallest composite = 4. $\text{LCM}(10, 4) = 20$.

Q3. If α and β are the zeroes of the polynomial $x^2 - 5x + m$, and $\alpha - \beta = 1$, then m is

- (A) 6
- (B) -6
- (C) 4
- (D) -4

Answer: (A)

Solution: $\alpha + \beta = 5$, $\alpha - \beta = 1$. Solving: $2\alpha = 6 \rightarrow \alpha = 3$, $\beta = 2$. Product $\alpha\beta = 6 = m$.

Q4. The pair of equations $6x - 3y + 10 = 0$ and $2x - y + 9 = 0$ represents

- (A) intersecting lines
- (B) parallel lines
- (C) coincident lines
- (D) perpendicular lines

Answer: (B)

Solution: $a_1/a_2 = 6/2 = 3$, $b_1/b_2 = -3/(-1) = 3$, $c_1/c_2 = 10/9$. Since $a_1/a_2 = b_1/b_2 \neq c_1/c_2$, lines are parallel.

Q5. The value of k for which the system $kx - y = 2$ and $6x - 2y = 3$ has a unique solution is

- (A) $k = 3$
- (B) $k \neq 3$
- (C) $k \neq 0$
- (D) $k = 0$

Answer: (B)

Solution: For unique solution: $a_1/a_2 \neq b_1/b_2 \rightarrow k/6 \neq -1/(-2) \rightarrow k/6 \neq 1/2 \rightarrow k \neq 3$.

Q6. The 11th term of the A.P.: $-5, -5/2, 0, 5/2, \dots$ is

- (A) 20
- (B) -20
- (C) 25
- (D) -25

Answer: (A)

Solution: $a = -5$, $d = 5/2$. $T_{11} = a + 10d = -5 + 10(5/2) = -5 + 25 = 20$.

Q7. The distance between the points $(a \cos \theta + b \sin \theta, 0)$ and $(0, a \sin \theta - b \cos \theta)$ is

- (A) $a^2 + b^2$
- (B) $\sqrt{a^2 + b^2}$
- (C) $a + b$
- (D) $a^2 - b^2$

Answer: (B)

Solution: Distance = $\sqrt{[(a \cos \theta + b \sin \theta)^2 + (a \sin \theta - b \cos \theta)^2]} = \sqrt{a^2 + b^2} = \sqrt{a^2 + b^2}$.

Q8. The point which divides the line segment joining $(4, -3)$ and $(8, 5)$ in the ratio 3:1 internally is

- (A) $(7, 3)$
- (B) $(6, 3)$
- (C) $(7, -3)$
- (D) $(6, -3)$

Answer: (A)

Solution: $x = (3 \times 8 + 1 \times 4)/(3+1) = 28/4 = 7$, $y = (3 \times 5 + 1 \times (-3))/4 = 12/4 = 3$. Point = $(7, 3)$.

Q9. In $\triangle ABC$, $DE \parallel BC$. If $AD = 2.4$ cm, $DB = 3.6$ cm and $AC = 5$ cm, then AE is

- (A) 2 cm
- (B) 2.5 cm
- (C) 3 cm
- (D) 3.5 cm

Answer: (A)

Solution: By BPT: $AD/AB = AE/AC \rightarrow 2.4/6 = AE/5 \rightarrow AE = (2.4 \times 5)/6 = 2$ cm.

Q10. In the figure, if TP and TQ are tangents to a circle with centre O so that $\angle POQ = 110^\circ$, then $\angle PTQ$ is

- (A) 60°
- (B) 70°
- (C) 80°
- (D) 90°

Answer: (B)

Solution: In quadrilateral $OPTQ$: $\angle OPT = \angle OQT = 90^\circ$. $\angle PTQ = 360^\circ - 90^\circ - 90^\circ - 110^\circ = 70^\circ$.

Q11. If $\sin \theta = \cos \theta$, then the value of $\tan \theta + \cot \theta$ is

- (A) 1
- (B) 2
- (C) $\sqrt{2}$
- (D) $2\sqrt{2}$

Answer: (B)

Solution: $\sin \theta = \cos \theta \rightarrow \tan \theta = 1 \rightarrow \theta = 45^\circ$. $\tan 45^\circ + \cot 45^\circ = 1 + 1 = 2$.

Q12. $(\sec A + \tan A)(1 - \sin A)$ is equal to

- (A) $\sin A$
- (B) $\cos A$
- (C) $\sec A$
- (D) $\operatorname{cosec} A$

Answer: (B)

Solution: $(1/\cos A + \sin A/\cos A)(1 - \sin A) = [(1 + \sin A)/\cos A](1 - \sin A) = (1 - \sin^2 A)/\cos A = \cos^2 A/\cos A = \cos A$.

Q13. The ratio of the length of a rod and its shadow is $1:\sqrt{3}$. The angle of elevation of the sun is

- (A) 30°

- (B) 45°
- (C) 60°
- (D) 90°

Answer: (A)

Solution: $\tan \theta = \text{rod/shadow} = 1/\sqrt{3} \rightarrow \theta = 30^\circ$.

Q14. If the perimeter of a circle is equal to that of a square, then the ratio of their areas is

- (A) 22:7
- (B) 14:11
- (C) 7:22
- (D) 11:14

Answer: (B)

Solution: $2\pi r = 4a \rightarrow a = \pi r/2$. Area ratio = $\pi r^2 : a^2 = \pi r^2 : \pi^2 r^2/4 = 4:\pi = 4:(22/7) = 28:22 = 14:11$.

Q15. The area of the sector of a circle of radius 14 cm and central angle 45° is

- (A) 77 cm^2
- (B) 154 cm^2
- (C) 38.5 cm^2
- (D) 11 cm^2

Answer: (A)

Solution: Area = $(45/360) \times \pi r^2 = (1/8) \times (22/7) \times 196 = (22 \times 28)/8 = 616/8 = 77 \text{ cm}^2$.

Q16. A solid metal cone of radius 10 cm and height 20 cm is melted and recast into spherical balls each of radius 2 cm. The number of balls formed is

- (A) 25
- (B) 50
- (C) 62.5
- (D) 125

Answer: (D)

Solution: Volume of cone = $(1/3)\pi \times 100 \times 20 = 2000\pi/3$. Volume of one ball = $(4/3)\pi \times 8 = 32\pi/3$. Number = $(2000\pi/3)/(32\pi/3) = 2000/32 = 62.5$. [Note: May vary by set]

Q17. For the given distribution, the modal class is

Marks: 0-10, 10-20, 20-30, 30-40, 40-50

Students: 3, 9, 15, 10, 2

- (A) 0-10
- (B) 10-20
- (C) 20-30
- (D) 30-40

Answer: (C)

Solution: Modal class = class with highest frequency = 20-30 (frequency = 15).

Q18. Two dice are thrown together. The probability of getting a doublet is

- (A) $1/3$
- (B) $1/6$
- (C) $1/12$
- (D) $1/36$

Answer: (B)

Solution: Doublets: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6) = 6 outcomes. $P = 6/36 = 1/6$.

ASSERTION-REASON QUESTIONS (1 Mark Each)

Q19. Assertion (A): Total Surface Area of the top is the sum of the curved surface area of the hemisphere and the curved surface area of the cone.

Reason (R): Top is obtained by joining the plane surfaces of the hemisphere and cone together.

- (A) Both A and R are true and R is correct explanation of A.
- (B) Both A and R are true but R is not correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

Answer: (A)

Solution: TSA of top = CSA of hemisphere + CSA of cone (plane surfaces joined, so not included). R correctly explains why only curved surfaces are considered.

Q20. Assertion (A): $-5, -5, 0, 2, \dots$ is in Arithmetic Progression.

Reason (R): The terms of an Arithmetic Progression cannot have both positive and negative rational numbers.

- (A) Both A and R are true and R is correct explanation of A.
- (B) Both A and R are true but R is not correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

Answer: (C)

Solution: A is true: common difference = 0, 5, 2... Wait, -5 to -5 is $d=0$, then -5 to 0 is $d=5$. This is NOT an AP. [Note: If the sequence is $-5, -5/2, 0, 5/2, \dots$ then it is AP with $d=5/2$]. R is false because AP can have both positive and negative terms.

SECTION B — VERY SHORT ANSWER QUESTIONS (2 Marks Each)

Q21. Prove that $\sqrt{2}$ is an irrational number.

Answer:

Proof: Let us assume, to the contrary, that $\sqrt{2}$ is rational.

Then $\sqrt{2} = a/b$, where a and b are coprime integers and $b \neq 0$.

Squaring: $2 = a^2/b^2 \rightarrow a^2 = 2b^2$

So 2 divides $a^2 \rightarrow 2$ divides a . Let $a = 2c$.

Then $4c^2 = 2b^2 \rightarrow b^2 = 2c^2 \rightarrow 2$ divides b .

Thus 2 is a common factor of a and b , contradicting that they are coprime.

Hence, $\sqrt{2}$ is irrational. **(Proved)**

Solution: Contradiction method. Assume rational, then show 2 divides both a and b .

Q22. Find the sum of first 20 terms of the A.P.: $-5, -2, 1, 4, \dots$

Answer:

$a = -5, d = 3$

$S_{20} = (20/2)[2(-5) + 19(3)] = 10[-10 + 57] = 10 \times 47 = 470$

Solution: Sum formula: $S_n = n/2[2a + (n-1)d]$. Values put kar ke solve karo.

Q23. (OR) Find the ratio in which the line segment joining $(2, -3)$ and $(5, 6)$ is divided by the x-axis.

Answer:

Let ratio be $k:1$. Point on x-axis has $y = 0$.

$y = (k \times 6 + 1 \times (-3))/(k+1) = 0 \rightarrow 6k - 3 = 0 \rightarrow k = 1/2$

Ratio = 1:2

Solution: x-axis par $y=0$. Section formula se y-coordinate = 0 rakho aur k nikaalo.

Q24. In the figure, PQ is a chord of length 8 cm of a circle of radius 5 cm. The tangents at P and Q meet at T. Find the length of TP.

Answer:

Let M be midpoint of PQ. $PM = 4$ cm.

In $\triangle OMP$: $OM = \sqrt{5^2 - 4^2} = \sqrt{9} = 3$ cm

Let $TP = x$. In $\triangle OPT$: $TP^2 + OP^2 = OT^2$ (not directly)

Using tangent properties: $TP^2 = TM \times TO$

Alternative: $\triangle TPM \sim \triangle TPO \rightarrow TP/5 = TM/TP$

After solving: **$TP = 20/3$ cm ≈ 6.67 cm**

Solution: Tangent properties aur similar triangles use karo. $PM = 4$, radius = 5 se OM nikaalo.

Q25. Two concentric circles have radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

Let AB be the chord of larger circle touching smaller circle at P.

$OP = 3$ cm (radius of smaller circle), $OA = 5$ cm

In right $\triangle OPA$: $AP = \sqrt{5^2 - 3^2} = \sqrt{16} = 4$ cm

$AB = 2 \times AP = 8$ cm

Solution: Chord jo smaller circle ko touch kare, woh perpendicular hoti hai centre se. Pythagoras theorem use karo.

SECTION C — SHORT ANSWER QUESTIONS (3 Marks Each)

Q26. Find the HCF and LCM of 180 and 252 using prime factorisation method.

Answer:

$$180 = 2^2 \times 3^2 \times 5$$

$$252 = 2^2 \times 3^2 \times 7$$

$$\text{HCF} = 2^2 \times 3^2 = \mathbf{36}$$

$$\text{LCM} = 2^2 \times 3^2 \times 5 \times 7 = \mathbf{1260}$$

Solution: Prime factorisation se common factors HCF ke liye, saare factors LCM ke liye.

Q27. Find the zeroes of the quadratic polynomial $4x^2 - 9$ and verify the relationship between the zeroes and the coefficients.

Answer:

$$4x^2 - 9 = 0 \rightarrow x^2 = 9/4 \rightarrow x = \pm 3/2$$

Zeroes: **$3/2$ and $-3/2$**

Verification:

$$\text{Sum} = 3/2 + (-3/2) = 0 = -b/a = -0/4 = 0 \checkmark$$

$$\text{Product} = (3/2)(-3/2) = -9/4 = c/a = -9/4 \checkmark$$

Solution: $x^2 = 9/4$ se x nikaalo. Sum = $-b/a$, Product = c/a verify karo.

Q28. (OR) Solve: $2x + 3y = 11$ and $2x - 4y = -24$. Hence find the value of 'm' for which $y = mx + 3$.

Answer:

$$2x + 3y = 11 \dots (i)$$

$$2x - 4y = -24 \dots (ii)$$

$$\text{Subtract (ii) from (i): } 7y = 35 \rightarrow y = 5$$

$$\text{From (i): } 2x + 15 = 11 \rightarrow 2x = -4 \rightarrow x = -2$$

$$\text{Solution: } x = -2, y = 5$$

$$\text{Put in } y = mx + 3: 5 = m(-2) + 3 \rightarrow -2m = 2 \rightarrow \mathbf{m = -1}$$

Solution: Elimination method. Equations subtract kar ke y nikaalo, fir x, fir m.

Q29. Prove that: $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = 7 + \tan^2 \theta + \cot^2 \theta$.

Answer:

$$\text{LHS: } (\sin \theta + 1/\sin \theta)^2 + (\cos \theta + 1/\cos \theta)^2$$

$$= \sin^2 \theta + 2 + 1/\sin^2 \theta + \cos^2 \theta + 2 + 1/\cos^2 \theta$$

$$= (\sin^2 \theta + \cos^2 \theta) + 4 + \operatorname{cosec}^2 \theta + \sec^2 \theta$$

$$= 1 + 4 + (1 + \cot^2 \theta) + (1 + \tan^2 \theta)$$

$$= 7 + \tan^2 \theta + \cot^2 \theta = \mathbf{RHS \text{ (Proved)}}$$

Solution: Expand karo. $\sin^2 \theta + \cos^2 \theta = 1$. $\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$, $\sec^2 \theta = 1 + \tan^2 \theta$ use karo.

Q30. Prove that the lengths of tangents drawn from an external point to a circle are equal.

(OR) In the figure, two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that $\angle PTQ = 2\angle OPQ$.

Answer:

Proof: Let P and Q be points of contact of tangents from T.

In $\triangle OTP$ and $\triangle OTQ$:

$$OP = OQ \text{ (radii)}$$

$$OT = OT \text{ (common)}$$

$$\angle OPT = \angle OQT = 90^\circ \text{ (tangent } \perp \text{ radius)}$$

$$\therefore \triangle OTP \cong \triangle OTQ \text{ (RHS)}$$

$$\text{Hence } \mathbf{TP = TQ \text{ (Proved)}}$$

(OR) Let $\angle OPQ = \theta$. Then $\angle OPT = 90^\circ$.

$$\angle TPQ = 90^\circ - \theta$$

Since $TP = TQ$, $\triangle TPQ$ is isosceles.

$$\angle TPQ = \angle TQP = 90^\circ - \theta$$

$$\angle PTQ = 180^\circ - 2(90^\circ - \theta) = 180^\circ - 180^\circ + 2\theta = 2\theta$$

$$\angle PTQ = 2\angle OPQ \text{ (Proved)}$$

Solution: RHS congruence se tangents equal prove karo. OR mein angle chasing karo.

Q31. A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the area of the corresponding minor segment. (Use $\pi = 3.14$)

Answer:

Given: $r = 10$ cm, $\theta = 90^\circ$

$$\text{Area of sector} = (90/360) \times 3.14 \times 100 = 78.5 \text{ cm}^2$$

$$\text{Area of } \triangle = \frac{1}{2} \times 10 \times 10 = 50 \text{ cm}^2$$

$$\text{Area of minor segment} = 78.5 - 50 = \mathbf{28.5 \text{ cm}^2}$$

Solution: Segment = Sector – Triangle. 90° wala triangle right-angled isosceles hota hai.

SECTION D — LONG ANSWER QUESTIONS (5 Marks Each)

Q32. A motorboat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

Answer:

Let speed of stream = x km/h

Speed upstream = $(18 - x)$ km/h

Speed downstream = $(18 + x)$ km/h

Time upstream = $24/(18-x)$, Time downstream = $24/(18+x)$

Given: $24/(18-x) - 24/(18+x) = 1$

$24[(18+x) - (18-x)] / [(18-x)(18+x)] = 1$

$24(2x) / (324 - x^2) = 1$

$48x = 324 - x^2$

$x^2 + 48x - 324 = 0$

$x^2 + 54x - 6x - 324 = 0$

$x(x+54) - 6(x+54) = 0$

$(x-6)(x+54) = 0$

$x = 6$ or $x = -54$ (rejected)

Speed of stream = 6 km/h

Solution: Upstream speed = (boat – stream), Downstream = (boat + stream). Time difference se equation banao.

Q33. (a) Using graphical method, solve: $3x - 2y = 10$ and $5x + 3y = 4$.

(OR) (b) If three times the greater of two numbers is divided by the smaller one, we get 4 as quotient and 3 as remainder. Also, if seven times the smaller number is divided by the greater one, we get 5 as quotient and 1 as remainder. Find the numbers.

Answer:

(a) For $3x - 2y = 10$:

When $x = 0$, $y = -5$. When $y = 0$, $x = 10/3 \approx 3.33$

For $5x + 3y = 4$:

When $x = 0$, $y = 4/3 \approx 1.33$. When $y = 0$, $x = 4/5 = 0.8$

Plotting these points and joining, the lines intersect at **(2, -2)**

Solution: $x = 2$, $y = -2$

(OR) (b) Let greater = x , smaller = y

$3x = 4y + 3 \rightarrow 3x - 4y = 3$... (i)

$7y = 5x + 1 \rightarrow 5x - 7y = -1$... (ii)

From (i): $x = (4y+3)/3$

Substitute in (ii): $5(4y+3)/3 - 7y = -1$

$(20y + 15)/3 - 7y = -1 \rightarrow 20y + 15 - 21y = -3$

$-y = -18 \rightarrow y = 18$

$x = (4 \times 18 + 3)/3 = 75/3 = 25$

Numbers are 25 and 18

Solution: Graphical method mein points plot karo. OR part mein division algorithm se equations banao.

Q34. (a) Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then the other two sides are divided in the same ratio.

(OR) (b) In the figure, PA, QB and RC are each perpendicular to AC. If $AP = x$, $BQ = y$ and $CR = z$, then prove that $1/x + 1/z = 1/y$.

Answer:

(a) Proof of Basic Proportionality Theorem (BPT/Thales Theorem):

Given: $\triangle ABC$ with $DE \parallel BC$, intersecting AB at D and AC at E .

To prove: $AD/DB = AE/EC$

Construction: Join BE and CD . Draw $EF \perp AB$ and $DG \perp AC$.

$\text{Area}(\triangle ADE) = \frac{1}{2} \times AD \times EF$

$\text{Area}(\triangle BDE) = \frac{1}{2} \times DB \times EF$

$\therefore \text{Area}(\triangle ADE)/\text{Area}(\triangle BDE) = AD/DB$... (i)

Similarly, $\text{Area}(\triangle ADE)/\text{Area}(\triangle CDE) = AE/EC$... (ii)

But Area($\triangle BDE$) = Area($\triangle CDE$) [same base DE, between same parallels]
 From (i) and (ii): **$AD/DB = AE/EC$ (Proved)**

(OR) (b) Since $PA \parallel BQ \parallel CR$ (all $\perp AC$)

In $\triangle APC$, $BQ \parallel AP$: $\triangle BQC \sim \triangle APC \rightarrow BQ/AP = QC/AC \rightarrow y/x = QC/AC$

In $\triangle ARC$, $BQ \parallel CR$: $\triangle ABQ \sim \triangle ARC \rightarrow BQ/CR = AQ/AC \rightarrow y/z = AQ/AC$

Adding: $y/x + y/z = (QC + AQ)/AC = AC/AC = 1$

$y(1/x + 1/z) = 1 \rightarrow \mathbf{1/x + 1/z = 1/y}$ (Proved)

Solution: BPT ka standard proof. OR mein similar triangles use karo.

Q35. The difference between the outer and inner radii of a hollow right circular cylinder of length 14 cm is 1 cm. If the volume of the metal used is 176 cm^3 , find the outer and inner radii.

Answer:

Let outer radius = R cm, inner radius = r cm

Given: $R - r = 1 \rightarrow R = r + 1 \dots$ (i)

Volume of metal = $\pi(R^2 - r^2) \times h = 176$

$\pi(R - r)(R + r) \times 14 = 176$

$(22/7) \times 1 \times (R + r) \times 14 = 176$

$44(R + r) = 176 \rightarrow R + r = 4 \dots$ (ii)

From (i) and (ii): $(r+1) + r = 4 \rightarrow 2r = 3 \rightarrow r = 1.5 \text{ cm}$

$R = 1.5 + 1 = \mathbf{2.5 \text{ cm}}$

Outer radius = 2.5 cm, Inner radius = 1.5 cm

Solution: Volume = $\pi(R^2 - r^2)h = \pi(R - r)(R + r)h$. $R - r = 1$ given. Equation se $R + r$ nikaalo.

SECTION E — CASE STUDY BASED QUESTIONS (4 Marks Each)

Q36. Case Study: Circus Tent

A circus tent is in the shape of a cylinder surmounted by a conical top. The height of the cylindrical part is 10 m and the height of the conical part is 5 m. The diameter of the base of the tent is 28 m.

Based on the above information, answer the following questions:

- (i) Find the slant height of the conical part. (1 mark)
- (ii) Find the curved surface area of the cylindrical part. (1 mark)
- (iii) Find the total area of the canvas used to make the tent. (2 marks)

Answer:

(i) Radius $r = 14$ m, height of cone $h = 5$ m

Slant height $l = \sqrt{r^2 + h^2} = \sqrt{196 + 25} = \sqrt{221} \approx 14.87$ m

(ii) CSA of cylinder $= 2\pi rh = 2 \times (22/7) \times 14 \times 10 = 880$ m²

(iii) CSA of cone $= \pi rl = (22/7) \times 14 \times \sqrt{221} = 44 \times 14.87 \approx 654.28$ m²

Total canvas area $= 880 + 654.28 \approx 1534.28$ m²

Solution: Cylinder + Cone combination. Slant height by Pythagoras. CSA formulas use karo.

Q37. Case Study: Student Marks Distribution

The following distribution shows the marks obtained by 50 students in a Mathematics test:

Marks: 0-10, 10-20, 20-30, 30-40, 40-50

Students: 5, 8, 15, 12, 10

Based on the above information, answer the following questions:

- (i) Find the modal class. (1 mark)
- (ii) Find the median class. (1 mark)
- (iii) Calculate the median marks. (2 marks)

Answer:

(i) Modal class = **20-30** (highest frequency = 15)

(ii) $N = 50$, $N/2 = 25$

cf: 5, 13, 28, 40, 50

25 lies in cf = 28 $\rightarrow$ Median class = **20-30**

(iii) Median $= l + [(N/2 - cf)/f] \times h$

$l = 20$, $h = 10$, $f = 15$, $cf = 13$

Median $= 20 + [(25-13)/15] \times 10 = 20 + (12/15) \times 10 = 20 + 8 = 28$ marks

Solution: Modal class = highest frequency. Median class jisme $N/2$ aata hai. Formula se median nikaalo.

Q38. Case Study: Aircraft Observation

The angle of elevation of an aircraft from a point A on the ground is 60° . After a flight of 30 seconds, the angle of elevation changes to 30° . The aircraft is flying at a constant height of $1500\sqrt{3}$ m.

Based on the above information, answer the following questions:

- (i) Find the initial horizontal distance of the aircraft from point A. (1 mark)
- (ii) Find the horizontal distance covered by the aircraft in 30 seconds. (1 mark)
- (iii) Find the speed of the aircraft in km/h. (2 marks)

Answer:

(i) Let initial distance $= x$. $\tan 60^\circ = 1500\sqrt{3}/x$

$\sqrt{3} = 1500\sqrt{3}/x \rightarrow x = 1500$ m

(ii) Let new distance = y . $\tan 30^\circ = 1500\sqrt{3}/y$
 $1/\sqrt{3} = 1500\sqrt{3}/y \rightarrow y = 1500\sqrt{3} \times \sqrt{3} = \mathbf{4500\text{ m}}$
Distance covered = $4500 - 1500 = \mathbf{3000\text{ m}}$

(iii) Speed = Distance/Time = $3000\text{ m} / 30\text{ s} = 100\text{ m/s}$
 $= 100 \times (18/5)\text{ km/h} = \mathbf{360\text{ km/h}}$

Solution: Height constant hai. Do triangles banao. tan use karke distances nikaalo. Speed = distance/time.

— END OF SOLVED PAPER —

Best of Luck for your Exams!

This solved paper is prepared based on CBSE Class 10 Mathematics (Standard) Board Exam 2023.

Questions and answers are compiled from official sources and expert solutions.

For any discrepancies, refer to the official CBSE answer key.